Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/19/2005
PI Name:	Ullrich, Robert Ph.D.		
Project Title:	NSCOR: Radiation Leukemogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bullrich@utmb.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	409-747-1935
Organization Name:	University of Texas Medical Branch		
PI Address 1:	301 University Blvd		
PI Address 2:	Comprehensive Cancer Center, MS 1048		
PI Web Page:			
City:	Galveston	**State:**	TX
Zip Code:	77555-5302	**Congressional District:**	14
Comments:	NOTE: PI moved to UTMB from Colorado State University in late 2008 (6/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	NSCOR 03-OBPR-02
Start Date:	10/01/2003	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bailey, Susan (Colorado State University) Bedford, Joel (Colorado State University) Belmont, John (Baylor College of Medicine) Borak, Thomas (Colorado State University) Cai, Wei-Wen (Baylor College of Medicine) Dingfelder, Michael (Carolina University) Ehrhart, E.J. (Colorado State University) Liber, Howard (Colorado State University) Nagarajan, Lalitha (U.T., M.D. Anderson. Cancer Center) Story, Michael (University of Texas, M. D. Anderson Cancer Center) Weil, Michael (U.T., M.D. Anderson. Cancer Center)		
Grant/Contract No.:	NAG9-1569		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of this program is to provide information required to develop a rational scientific basis for estimation of risks for leukemogenesis in humans from exposure to radiation during space flight. This will be accomplished using several coordinated research activities. A key activity will be to develop quantitative data on the relative leukemogenic effectiveness of specific HZE particles and of high-energy protons compared with gamma rays in a mouse model of radiation-induced AML. Extrapolation of risks from this mouse model to risks in humans requires additional information, and a major component of research activities in this program are designed to provide such information. A major research effort of this program is focused on studies of cellular and molecular mechanisms and processes involved in radiation-induced AML. These studies will not only improve the basic understanding of the pathogenesis of AML but will also provide information on cellular and molecular biomarkers that may be useful in predicting individuals at increased risk. These activities will focus on the use of molecular and molecular cytogenetic approaches, the development of new model systems, and comparative analyses of murine and human AML to facilitate the translation of results obtained in mice to human AMLs. Problems of extrapolation are not limited to biological factors. There are clear dosimetric differences that also must be considered when extrapolating data obtained from mouse studies to human risks. To address these issues this project also has an important physics and dosimetry component that includes modeling of dose to target cells in murine and human bone marrow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Not Available
	Project 1. Induction of AML in CBA/CaJ mice (project leader, Dr. Robert. L. Ullrich). This project is designed to quantitatively compare the leukemogenic effects of irradiation with gamma rays, select HZE particles, and protons using the CBA/CaJ murine model of AML. These experiments are designed to provide sufficient quantitative data to permit the determination of their relative effectiveness by comparing slope constants obtained over specified dose ranges rather than fully defining the dose response relationships for each of these radiations. Over this project period irradiations of mice for the acute HZE and gamma ray irradiations were completed. We are finishing dosimetry for the low dose rate (LDR) irradiation groups and will begin irradiations in the fall. In all almost 1100 mice (extra mice were added to allow for loss during travel post irradiation) have been entered into the project 1 study with HZE particles and approximately 200 are over 400 days post irradiation. This is the time at which we should expect to begin to see AML. Indeed, we are beginning to find mice with AML in these groups. Likewise for the acute gamma ray group, all 800 mice have been irradiated and we are beginning to see the development of AML. Project 2: Quantification and characterization of specific cytogenetic changes associated with initiation, progression, and development of AML (Project leader, J.S. Bedford). Determination of the relative biological effectiveness (RBE) of select HZE radiations relative to gamma-radiation over specified dose ranges for induction of AML in CBA/CaJ mice is well underway as outlined under project 1 above. In parallel, groups of mice exposed to the same radiation doses are being studied to follow the relative effectiveness of these radiations for the production of chromosomal abnormalities that are known to be associated with radiation induced AML in these mice, as well as the relative effectiveness for production of the same chromosome abnormalities in another strain of mice, C57/BL6, shown in previous studies to be more resistant to radiation induction of AML. Virtually all (at least 95%) of the cells of leukemias that do develop in the irradiated CBA mice contain a deletion of the PU.1 gene located on chromosome 2. The experimental plan we have been carrying out during the past year and a half involves measurement of the loss of a small segment of mouse chromosome 2 where this gene resides, in bone marrow cells sampled 24 hours, 1 month, and 1 year after irradiation in the two mouse strains for the different doses and different radiations. To this end, we have isolated and fluorescently labeled BAC clones containing the PU.1 gene; in particular we chose BAC RP23-263H8, which is197 kb in size. It is located on chromosome 2 in region 2E1, mapping at 49.3 cM or (by physical mapping) between 90,840 - 91,037 kb from the centromere. We verified that the correct probe (BAC) was isolated utilizing a PCR method with unique primers for this BAC, and the labeled BAC was then used as a FISH probe on metaphase cells from mouse bone marrow. The result from PCR verification confirms our BAC was indeed located on chromosome 2 and in the expected location. Project 3A: Proximity of Radiation Induced Deletion Breakpoints in Mouse Chromosome 2 in Relation to Myeloid Tissue-Specific Chromatin Structure. (Joel Bedford, Yuanlin Peng, and David Maragnon) Nikiforova and co-workers reported in 2000 that two regions on human chromosome 10 are associated with the breakpoints producing an inversion that involving fusion of part of the RET gene with the H4 gene located some 30 Mb away. This inversion occurs in an appreciable number of patients with thyroid cancer in patients who had been exposed to radiation as a result to the Chernobyl accident. Using BAC probes containing these regions, their report showed that the breakpoint regions on chromosome 10 were much closer together in normal thyroid cells than predicted, and closer than actually measured in breast epithelial cells, i.e., another tissue. Since the data of Bouffler and Cox and co-workers at Harwell in the U.K. had demonstrated both a proximal and a distal cluster of breakpoints around the PU.1 gene involved in the radiation induced leukemias of CBA mice, we hypothesized that a similar proximity of these breakpoints may be involved in the bone marrow cell precursors of the leukemias. To test this hypothesis, we isolated additional BAC probes reported in the breakpoint regions. We hasten to add that this situation is not the exact equivalent of a precise inversion but involves a cluster, so we had planned to look at several regions in the cluster. The first step was to isolate and label probes and confirm they were indeed located on chromosome 2 in about the expected location. The first preliminary result shows a metaphase in which the PU.1 probe, already established to be located on chromosome 2, is located between two green fluorescent signals on the same chromosome. One of the green (FITC) labeled probes was from BAC RP23-1011 located at 36.8cM on chromosome 2 and proximal (nearer the centromere) to the PU.1-containing probe. The other green labeled probe was from BAC RP23-34E24 located at 56.8 cM on chromosome 2 and distal to the PU.1-containing probe. There was some indication of a more diffuse labeling and we attribute this to slightly suboptimal hybridization of these two breakpoint cluster probes at this time. The probe size after labeling by nick-translation was sub-optimal. Nevertheless, results demonstrated that the probes we have isolated are on the expected chromosome at the expected position. Project 3B: CGH Signatures of Spontaneous and Radiation-Induced AMLs (Project Leader, W-W Cai). Our project involves using whole genome BAC arrays to analyze mouse leukemia samples generated in the project. We are ready to accept samples for analysis and have performed some analysis for AML samples from Simon Boufler’s lab in UK. In

Task Progress:

the meantime we continue to improve the quality and resolution of our mouse whole genome BAC arrays. We have recently selected a new set of tiling path clones with unique sequences at both ends. This will remove any potential chimeric clones and the map information for clone set will be reliable and relatively stable. The new set contains close to 22,000 BAC clones covering the genome in contigs with less than 300 gaps, most of which are located in the unclonable regions. The Y chromosome is also covered in this clones set. We have finished DNA preparation for the new clone set and high quality new arrays will be printed in the next two months.

Another significant development is our success in development of a new procedure for making Cot I blocking DNA. The homemade Cot I DNA works as well as the commercial Cot I DNA but the cost is significantly reduced. We are currently evaluating an automatic image quantitation software package. We expect that implementation of this package will greatly increase our sample throughput.

Project 3C: HZE Induced Chromosomal Instabilities and Complex Aberration Biomarker Stability in the Clonal Progeny of Mouse Myeloid Cells Irradiated and Assayed in vivo. (Project Leaders: S.M. Bailey and J.S. Bedford). We have carried out some preliminary studies to perfect the tail vein injections and determine the optimum dose to use for bone marrow ablation, but have not isolated HZE surviving bone marrow clonal cell populations for cytogenetics analysis.

Project 3D: Gene Expression in AML (Project Leader, M.D. Story) . At this time we await samples for processing. We have worked out the logistics associated with shipping samples and tested RNA quality as a result of shipping small tumor samples in RNAzol B. RNA quality and quantity are more than sufficient for array analysis. All components for array hybridization and data analysis are in place.

Project 4. Molecular and cytogenetic targets in murine and human AML. (Project leaders Drs. Joel Bedford and Robert Ullrich, Lalitha Nagarajan, and John Belmont).

4a. Common molecular and cytogenetic targets in murine and human AML (Project leaders Drs. Joel Bedford and Lalitha Nagarajan) Deletions of chromosomes 5q, 7q, 11q, 17p and 20q are recurrent anomalies in human acute myelogenous leukemia, secondary to radiation and or alkylating agent therapy. As part of the NSCOR, our focus in the past eighteen months has been to characterize deletion 7q. Based on the completed genomic sequence of chromosome 7, we have generated a transcript map of the critical region of loss (between the D7S525 and D7S2502 loci). Notably, a recently characterized tumor suppressor gene DOCK4 and an evolutionarily conserved zinc finger gene ZNF277 localize to this interval, head to head, within <0.5 kb of each other. A paper on this on recently been accepted for publication (Hong Liang, Patricia D. Castro, Jin Ma, and Lalitha Nagarajan. Finer Delineation and Transcript Map of the 7q31 Locus Deleted in Myeloid Neoplasms. Cancer Genet. Cytogenetics, in press). The reagents generated in this study will be valuable in elucidating the role of loss 7q31 loci in the pathogenesis AML. We will be interested in evaluating these genes in the murine heavy ion induced AML models that are being developed in other parts of the consortium.

Project 4b. Development of a mouse/human hybrid model of human leukemogenesis. (Project leader, Dr. John W. Belmont). These studies were designed to determine whether hematopoietic cells transplanted from the bone marrow of CBA strain mice to irradiated NOD/LtSz/Rag1null mice remain susceptible to radiation-induced AML and determine if transplantation affects the latency. CBA/CaJ and NOD/LtSzRag1null mice were obtained from Jackson laboratory (Bar Harbor, Maine), and colonies were established and expanded in the Baylor College of Medicine Center for Comparative Medicine Barrier facility. Currently, we are maintaining 20 breeding cages (one male and two female per cage) for the NOD/Rag strain and 12 breeding cages for the CBA/CaJ strain to have the sufficient number of age-appropriate animals to perform the experiments. The NOD/Rag strain showed particularly slow breeding habits.

Serial experiments have been performed to estimate the minimum number of CBA-CaJ Sca1+cKit+Lin- cells to be transplanted in NOD/Rag mice in order to obtain engraftment. Usually, but for different strains of mice, that number ranges from 1000 to 100,000 Sca1+cKit+ Lin-cells. Previous experience indicated that 30-50 million cells could be harvested from a murine BM (four bones collection) and that Sca1+cKit+Lin- constitute about 0.5% (about 100,000-200,000 cells/mouse). The Sca1+cKit+Lin- sorting has been optimized. Briefly, CBA/CaJ bone marrow mononuclear cells are stained with the BD Pharmigen mouse lineage panel, which contains biotinylated monoclonal antibodies to mouse CD3e, CD11b (Mac-1), CD45-B220, Ly-6G (Gr-1), and Ly-6c, and TER-119/Erythoroid cells (BD Pharmigen, Cat. No 559971). In a second step, cells are incubated with Avidin-FITC (BD Pharmigen , Cat. No. 554057), and with PE-Anti-mouse Ly-6A/E (Sca-1; Caltag Code No. MSCA04-3), and PE-Rat-Anti-CD117 (cKit; Caltag RM6204), and eventually sorted. We are recovering an average of 0.1% Sca-1+cKit+Lin- cells/mouse.

Five groups of two to ten NOD/Rag mice have received 25,000 to 50,000 CBA/CaJ Sca-1+cKit+Lin- cells by intra-tail vein injections. The recipient mice were irradiated immediately before the transplants with 4 Gy (1 Gy/min) in a 137Cs small animal irradiator. Engraftment of CBA cells in the host mice has been monitored from the third week post-transplantation by flow cytometric analysis for the marker H2-K specific of the CBA strain (NOD/LtSz/Rag1null carry the marker H2-D) from the murine peripheral blood. The technique has been optimized to reduce the level of background due to the monoclonal antibody capturing by the Fc receptor on the host cells. Briefly, NOD/Rag-transplanted-CBA/CaJ peripheral blood cells are preincubated with purified Rat anti-mouse CD16/CD32 (Fcg III/II Receptor; BD Pharmigen Cat. No. 553141) prior of the H2-K staining (BD Pharmigen Cat. No. 553592; FITC-labeled). We have observed an engraftment ranging from 2% to 16% in five NOD/Rag mice which received 50,000 CBA/CaJ cells. No engraftment was found in the mice which received 25,000 donor cells. CBA/CaJ positive-NOD/Rag mice were irradiated with 3Gy as established by the experimental procedure.

Presentations: Invited presentation at 3rd International Workshop on Space Radiation and 15th Annual NASA Space Radiation Health Investigator's Workshop, May 16-20, 2004: Modulation of genetic effects by RNA interference of NHEJ. S.M. Bailey, J.S. Bedford and H.L. Liber

Invited presentation at NASA Bioastronautics Investigators' Workshop, Galveston TX, January 10-12, 2005: DNA-PK knockdown by siRNA has differential quantitative effects on telomere dysfunction and mutagenesis in human lymphoblasts treated with gamma-rays or HZE particles. Qinming Zhang, Eli S. Williams, Joel S. Bedford, Howard L. Liber and Susan M. Bailey

Invited presentation at 16th Annual Space Radiation Health Investigator's Workshop, Port Jefferson, NY, May 15-18, 2005: Modulation of Genetic Effects by RNA interference of NHEJ. Susan M. Bailey, Joel S. Bedford and Howard L. Liber

Bibliography Type:	Description: (Last Updated: 06/10/2025)
Abstracts for Journals and Proceedings	Bailey SM, Peng Y, Zhang Q, Bedford JS, Liber H, Ullrich RL, Goodwin EH. "Dysfunctional mammalian telomeres join to double-strand breaks." Twelfth International Congress of Radiation Research. Brisbane, Australia, 17 – 22 August 2003. Twelfth International Congress of Radiation Research. Brisbane, Australia, 17 – 22 August 2003. , Aug-2003
Articles in Peer-reviewed Journals	Zhang Q, Williams ES, Askin KF, Peng Y, Bedford JS, Liber HL, Bailey SM. "Suppression of DNA-PK by RNAi has different quantitative effects on telomere dysfunction and mutagenesis in human lymphoblasts treated with gamma rays or HZE particles." Radiat Res. in press July 2005 , Jul-2005
Articles in Peer-reviewed Journals	Zhang Y, Lim CU, Williams ES, Zhou J, Zhang Q, Fox MH, Bailey SM, Liber HL. "NBS1 knockdown by small interfering RNA increases ionizing radiation mutagenesis and telomere association in human cells." Cancer Res. 2005 Jul 1;65(13):5544-53. PMID: 15994926 , Jul-2005
Articles in Peer-reviewed Journals	Liang H, Castro PD, Ma J, Nagarajan L. "Finer delineation and transcript map of the 7q31 locus deleted in myeloid neoplasms." Cancer Genet Cytogenet. in press, July 2005 , Jul-2005